Can the LeHome $\pi_{0.5}$ teacher grasp on demand? A start-state test in its own simulation

WorldFold, WatAI (Roy Sulaiman)
measurement run and write-up by Claude, 2026-10-19

Abstract

The teacher cannot be prompted (its only "instruction" is the garment's name), so to test the grasp sub-skill on its own we freeze the simulation a few steps before each grasp the policy made during ordinary folds, hand the policy that state cold, and score what it does in 60 steps. We do this for 60 restored states from 20 folding episodes in the teacher's own physics, once as recorded and once with the garment moved 3 cm and turned 15°. Restored as-is, 67% (68/102) of the policy's jaw closes pinched cloth and 76% (52/68) of those lifted it at least 4 cm; with the garment moved, 44% (48/109) and 60% (29/48) (Fisher $p = 0.07$ on the lift share). Counted over all attempts, the cloth ended up lifted in 51% (52/102) of the as-is attempts and 27% (29/109) of the moved ones ($p < 0.001$), against 78% (67/86) inside the teacher's own folds. Every restored episode produced at least one jaw close. The weights do hold a grasp, but about half of what looks like grasping in a fold is trajectory memory: moved 3 cm, the jaw closes where the cloth used to be.

1 Why this test

Run A scored the grasps the teacher chose while folding; a fold score cannot say whether the policy *has* a grasp skill, because its training physics lets it fold while dragging. The direct question is: given a moment that calls for a grasp, does it grasp, and does it still grasp when the cloth is not exactly where its memory of the fold would put it? Moving the garment separates a grasp skill from a replayed trajectory.

2 Method

Harvest. 20 folding episodes (the Run A setting: LeHome's shipped physics, stock arm, round jaws, Ilia's harness), saving the full physics state (cloth vertices and velocities, garment pose, both arms' joints) every 5 policy steps, with the grasp logger running. **Selection.** For every logged jaw close that was a grasp attempt, the snapshot 8–20 steps before the close (nearest to 10), i.e. with the jaw still open and approaching; at most three per garment. **Restore.** At each test episode, after the harness's normal reset and settle, the snapshot is written back exactly (Ilia's `restore_physics_state`) and the simulation settles 3 steps; in the moved condition the garment pose is first shifted 3 cm in a random direction and rotated $\pm 15°$ about its own centre, with the arms left where they were. The policy then runs cold (its action history reset) for at most 60 steps. **Scoring.** The same logger as Run A: pinched (at least five vertices in the pad box at the close), lifted (the pinched cloth rose $\geq$4 cm with the jaw), up at release (still $\geq$2 cm up when the jaw opened), held (the strict 2 cm rule, which Run A showed measures sag in this physics), layers. Episodes in which no jaw closed within the step budget are counted separately.

3 Results

Condition	Episodes	No close	Attempts	Pinched	Lifted ≥4 cm	Up at release	Held (2 cm)	Layers	First close (step)
Baseline (full folds)	20	0	86	78% (67/86)	100% (67/67)	55% (37/67)	4% (3/67)	3.0	149.0
Restored as-is	60	0	102	67% (68/102)	76% (52/68)	37% (25/68)	10% (7/68)	2.0	10.0
Restored, shifted 3 cm + rotated 15°	60	0	109	44% (48/109)	60% (29/48)	31% (15/48)	17% (8/48)	2.0	12.0
Restored as-is, 5 steps, then moved 3 cm + 15°	60	0	105	37% (39/105)	44% (17/39)	31% (12/39)	23% (9/39)	2.0	11.0

Table 1: The conditions. In the full folds an empty close at the episode's end is the arm parking and is not an attempt; in the 60-step restored episodes every empty close is a missed attempt unless made in the last 8 steps (no time to pinch). First close is the median policy step of the first attempt.

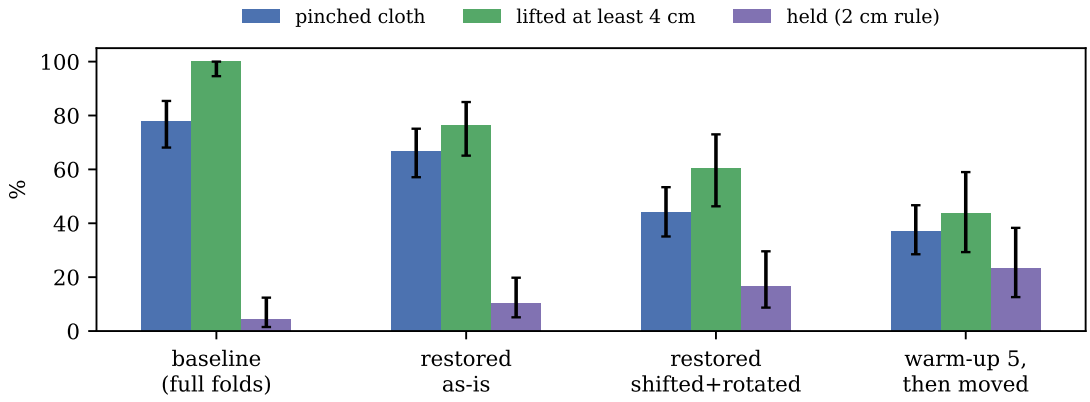


Figure 1: Pinched, lifted and held shares per condition with 95% Wilson intervals.

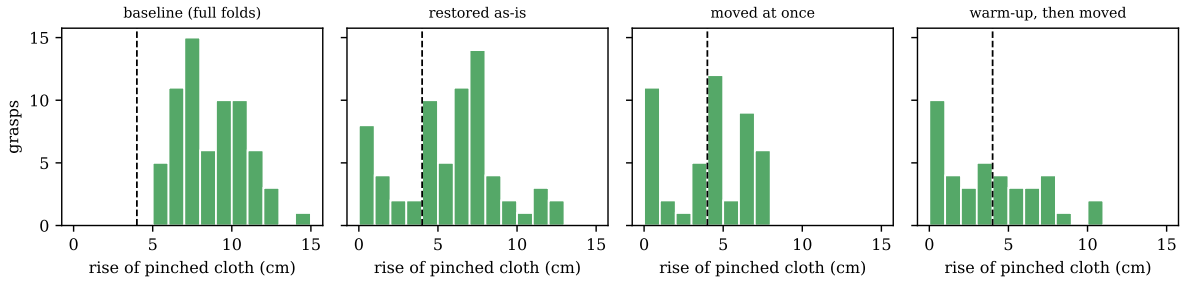


Figure 2: How far the pinched cloth rose with the jaw, per condition (dashed: the 4 cm lifted threshold).

Reading. As-is vs moved: pinched share 67% (68/102) vs 44% (48/109) (Fisher $p = 0.001$); lifted share among pinched 76% (52/68) vs 60% (29/48) ($p = 0.07$); lifted over all attempts 51% (52/102) vs 27% (29/109). A policy that only replays its fold would keep closing where the cloth used to be and miss after the move; a policy with a grasp skill re-aims. The move costs mainly the *pinch*: the jaw closes on air or on too little cloth far more often, while a grasp that does catch cloth still lifts it 60% of the time. In the warm-up variant (restored as recorded, 5 policy steps, then the same move) the cloth ended up lifted in 16% (17/105) of attempts (pinched 37% (39/105), lifted 44% (17/39) of pinched), against 27% (29/109) when moved before the policy saw the state (Fisher $p = 0.07$). By type, lifted over attempts: long-sleeve tops 67% (12/18) as-is vs 4% (1/24) moved; short-sleeve tops 10% (3/29) as-is vs 20% (6/30) moved; long pants 38% (10/26) as-is vs 23% (6/26) moved; shorts 93% (27/29) as-is vs 55% (16/29) moved. Long-sleeve tops collapse when moved and shorts barely notice, so the "skill" is strongest where the garment is small and compact and weakest where the target is a sleeve. On demand the policy also takes fewer layers (median 2 vs 3 in the folds), and

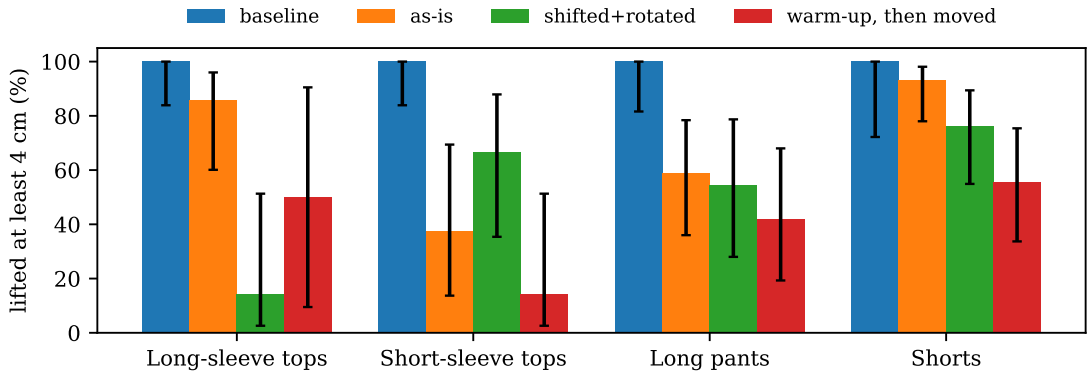


Figure 3: Lifted share by garment type and condition.

among moved grasps 21 of 48 caught a single layer. The per-garment table (Appendix A) shows which garments it re-aims on.

4 What this means

There is a grasp in the weights: handed a grasp moment cold, the teacher catches and lifts cloth in half of its attempts, and a quarter after the garment is moved by an amount a real table setup varies by. That is enough to RL-fine-tune from, and far more than a policy started from scratch would have; it is not a reliable sub-skill. The warm-up variant rules out the cold start as the cause of the drop: a policy that has already committed to its approach and then has the garment moved under it does no better (16% (17/105) of attempts), so what is missing is re-aiming from what the cameras show, not warm-up. The start states, the reward (the lift measure) and the baseline rates for that fine-tune are all produced by this test, so the next step needs no new machinery: roll out from these states, keep the grasps that lifted, update the teacher, re-run this test. The same mechanism serves the other sub-skills once scorers for placing and dragging exist.

5 Caveats

1. The policy starts cold from the restored state (no action history); the original grasp was made mid-chunk. A cold start can cost it a few steps, which is why the first-close step is reported.
2. The states are the teacher's own pre-grasp moments; this tests grasping where it already intended to grasp, moved by 3 cm / 15°. It does not test grasping an arbitrary point.
3. Same physics as Run A: the 2 cm held rule measures sag; the lift measures are the grasp result.
4. Three states per garment, 4 types: per-type numbers have wide intervals.
5. "Up at release" is not a success criterion: a grasp that places the cloth ends low by design. It is reported because it is what the attach contact model in the corrected physics will have to reproduce.

A Per garment

Garment	Condition	Episodes	Attempts	Pinched	Lifted ≥4 cm	Layers
Top_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	2	3 3 3
Top_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	4	1	1	1
Top_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	0	0	
Top_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	3	1 1 3
Top_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	0	1 1 1
Top_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	0	0	
Top_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	2	2 3 3
Top_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	1	0	2
Top_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	1	1	1
Top_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	3	1 2 3
Top_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	4	1	0	1
Top_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	1
Top_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	2	2	2 2
Top_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	2
Top_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	0	0	
Top_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	3
Top_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	2
Top_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	1	0	1
Top_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	2	1	1 2
Top_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	3
Top_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	2
Top_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	1	2
Top_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	1	2
Top_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	1	1 1 4
Top_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	1	1 1 4
Top_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	3	1 2 4
Top_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	4
Top_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	1	0	2
Top_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	2	1 1 5
Top_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	0	6
Pant_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	2	1 1 2
Pant_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	2	0	1 1
Pant_Long_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	7	3	0	1 2 4
Pant_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	5	1	1 2 2 2 4
Pant_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	4	0	0	
Pant_Long_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	2	0	1 2
Pant_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	5	3	1 1 2 3 3
Pant_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	3	1 1 4
Pant_Long_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	2	1 1 1
Pant_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	1	1	2
Pant_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	3	3	0	2 2 2
Pant_Long_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	2	1	0	1
Pant_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	3	1 1 3
Pant_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	3	1 2 4
Pant_Long_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	3	3	2 4 6
Pant_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	6	6	2 2 2 4 4 6
Pant_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	4	3	1 1 4 8
Pant_Short_Seen_0	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	4	1	1 1 1 2
Pant_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	6	6	2 2 2 2 2 3
Pant_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	5	4	1 2 2 5 6
Pant_Short_Seen_1	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	4	2	2 2 2 2
Pant_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	5	4	2 2 2 4 4
Pant_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	2	0	1 1
Pant_Short_Seen_2	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	2	1	1 1
Pant_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	6	6	1 2 2 5 5 5
Pant_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	5	5	1 1 1 2 2
Pant_Short_Seen_3	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	4	2	2 2 2 2
Pant_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	6	6	5	1 1 2 2 2 2
Pant_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	5	5	4	2 2 2 4 4
Pant_Short_Seen_4	'asis': 'as-is', 'pert': 'shifted+rotated', 'warm': 'warm-up, then moved'[k]	3	4	4	4	1 1 1 3

Table 2: Restore-driven episodes per garment and condition.